

Milky Way visible from 13% of Malta?

*Testing a University of Malta study of night-sky
brightness, and how it was reported*

THE CLAIM UNDER REVIEW

***“...87% of the area registering a NSB < 20.39 (Bortle Class 5
or higher) ... with the Milky Way being visible for only
12.8% of the area.”***

Caruana et al., Journal of Environmental Management, 2020 (University of Malta).
Reported in the news as “Milky Way only visible from 13% of Malta”.

VERDICT: SUPPORTED

Accurately reports a peer-reviewed measurement from 2017–2019

Version 1.0 · 5 October 2026

Status: no right of reply needed

Prepared from published research.

Repository: github.com/leandergrech/Mizien

MIŻIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

A University of Malta team measured the brightness of the night sky across the islands and reported, in the *Journal of Environmental Management* (2020), “**87% of the area registering a NSB < 20.39 mag_{SQM}/arcsec² (Bortle Class 5 or higher) ... with the Milky Way being visible for only 12.8% of the area**”. News reports rounded this to “13%”. We checked that the summary matches the study and that the study’s numbers hold together.

- 1

The summary matches the study. The paper gives 12.8% (Bortle class 4 threshold) or 13.5% (the world atlas threshold); “about 13%” is fair. 87% is the share of land in Bortle classes 5–9, as stated.
- 2

The study’s numbers hold together and agree with the world atlas. Weighting the island results by land area gives 11.6% (the paper weights its 347 survey cells, giving 12.8%). The 2016 world atlas of night-sky brightness gave 11% for Malta.
- 3

The threshold matters. On the paper’s own sensitivity test, the visible share is 6.3% to 25.9% depending on the brightness threshold chosen. “About 13%” uses a standard, stated threshold.
- 4

The data are from 2017–2019. Measurements are from 2017–2019. On the island of Malta the class 4 area fell from 3.9% to 1.6% between the two surveys. We found no newer survey.
- 5

Verdict: supported (high confidence). The statement reports a peer-reviewed measurement accurately.

THIS CLAIM

SUPPORTED

LARGELY SUPPORTED

NOT SUBSTANTIATED

MISLEADING

CONTRADICTED

12.8%

Area where the Milky Way is visible, 2017/18 (Bortle class 4)

87%

Area in Bortle class 5 or brighter

11%

The same measure in the 2016 world atlas

1.6%

Class 4 area on the island of Malta in 2018/19 (3.9% a year earlier)

WHAT WOULD MOVE THE VERDICT UP

None: Supported is the top of the scale.

WHAT WOULD MOVE IT DOWN

A newer survey showing a different share, or a finding that the study’s method misclassifies large areas.

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1 The claim and what we could verify

The claim record came from a MaltaToday headline, “Milky Way only visible from 13% of Malta”, which summarised the study. Following our rule to quote the speaker, not the headline, we read the study itself: Caruana, Vella, Spiteri, Nolle, Fenech and Aquilina (2020), “A photometric mapping of the night sky brightness of the Maltese islands”, *Journal of Environmental Management* 261:110196 [1], in its accepted version on arXiv [2]. The quoted wording is the paper’s abstract.

Source	What it says	Our access	Status
Caruana et al. (2020), J. Environ. Manage. [1, 2]	“87% of the area registering a NSB < 20.39 ... with the Milky Way being visible for only 12.8% of the area”; 13.5% on the world atlas threshold.	Accepted manuscript read in full (arXiv).	The claim
Falchi et al. (2016), Science Advances [3]	World atlas of artificial night-sky brightness: 11% for Malta (as cited by Caruana et al.).	Crossref verified; figure as cited ♦.	Comparison
MaltaToday [4]	“Milky Way only visible from 13% of Malta”.	Locator only (site refuses automated access).	Locator

2 Method

Question. Does the statement report the study accurately, and are the study’s results consistent?

Evidence. We transcribed the study’s Table 1 (share of land area in each Bortle class, by island) into *data/cc-053/caruana2020_table1.csv* and recomputed the archipelago-wide shares by weighting each island by its land area from OpenStreetMap (*tools/cc-053-report/calc.py*; outputs in *data/cc-053/checks.csv*). We compared the result with the paper’s own comparison to the 2016 world atlas and with its threshold sensitivity test.

Grades. Caruana et al. (2020): B (observational measurement with a stated method and comparison). **Verdicts** follow the five-point scale in Appendix A.

3

What the data show



Bortle classes from Sky Quality Meter readings in 347 one-km cells. Source: Caruana et al. (2020) Table 1.

Figure 1. Share of land area in each Bortle class by island (Caruana et al. 2020, Table 1).

No part of the islands reached Bortle class 3 or darker. Class 4, where the Milky Way is clearly visible, covered 3.9% of Malta, 37.2% of Gozo and 83.3% of Comino in 2017/18, 12.8% of the islands overall. Weighting by land area gives 11.6%; the paper weights its one-kilometre survey cells, which include coastal cells, so the darker coasts count for slightly more. On the island of Malta, the second survey (2018/19) found class 4 had shrunk to 1.6%.

4

Where the evidence points different ways

Is “13%” the right number?

YES, ON A STATED THRESHOLD

FOR

12.8% on the Bortle class 4 threshold, 13.5% on the world atlas threshold; the world atlas itself gave 11%.

CAVEATS

With a threshold of 20.6 mag/arcsec² the share is 6.3%; with 20.0 it is 25.9%. The measurements are from 2017–2019.

WHY THEY DIFFER, AND WHAT IT MEANS HERE

“About 13%” is accurate for the thresholds the study uses. It describes conditions in 2017–2019, which may since have changed.

5 Testing the claim

Sub-claim	Said by	What the evidence shows	Rating
A. The Milky Way is visible from only about 13% of the islands	UM researchers [1]	12.8% (class 4) or 13.5% (atlas threshold); 11.6% area-weighted; world atlas 11%.	ACCURATE
B. 87% of the area has high light pollution	UM researchers [1]	87.2% of the area in Bortle class 5 or brighter, 2017/18.	ACCURATE

6 Verdict and requests for evidence

VERDICT

SUPPORTED

The statement reports a peer-reviewed measurement accurately. Confidence: high.

Why. Both figures are the study's own, the study's parts add up, and an independent atlas agrees. **What this verdict does not say.** It does not describe the sky in 2026: the survey has not been repeated nationally since 2019.

Evidence we are asking for

- ☐ From the University of Malta team or ERA: any repeat survey of night-sky brightness since 2019.
- ☐ From ERA and the Planning Authority: the status of national lighting guidelines.

RIGHT OF REPLY

Not needed for this verdict (maintainer rule of 5 October 2026).

7 Limitations

- We read the accepted manuscript on arXiv; the published version may differ slightly in wording.
- The world atlas figure (11%) is as reported by Caruana et al.; we did not recompute it.
- Sky Quality Meter readings vary with season, cloud and the Milky Way's position; the study discusses these.

References

- [1] Caruana J., Vella R., Spiteri D., Nolle M., Fenech S. & Aquilina N.J. (2020). A photometric mapping of the night sky brightness of the Maltese islands. *Journal of Environmental Management* 261:110196. doi:10.1016/j.jenvman.2020.110196. <https://doi.org/10.1016/j.jenvman.2020.110196>
- [2] Same, accepted manuscript. arXiv:2002.04435. <https://arxiv.org/abs/2002.04435>
- [3] Falchi F., Cinzano P., Duriscoe D., Kyba C.C.M. et al. (2016). The new world atlas of artificial night sky brightness. *Science Advances* 2(6):e1600377. doi:10.1126/sciadv.1600377. <https://doi.org/10.1126/sciadv.1600377>
- [4] MaltaToday. Milky Way only visible from 13% of Malta (locator). https://www.maltatoday.com.mt/news/national/100416/milky_way_only_visible_from_13_of_malta
- [5] Miżien. Data and calculation: data/cc-053/; tools/cc-053-report/calc.py.

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. ♦ marks a source known only second-hand.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.
- **Offer a right of reply** where a check finds a claim Not substantiated, Misleading or Contradicted (or a pledge Not measurable, Off track or Missed), before wider circulation, and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	5 Oct 2026	First issue. Verdict Supported (high confidence); no right of reply needed.